Chapter 23 Integration and Hormonal Regulation of Mammalian Metabolism

Multiple Choice Questions

1. The radioimmunossay (RIA) is based on competition of unlabeled and radiolabeled:

   A) antibodies for binding to a hormone.
   B) antibodies for binding to a receptor.
   C) hormone for binding to a receptor.
   D) hormone for binding to an antibody.
   E) receptor for binding to a hormone.

Hormones: diverse structures for diverse functions
Pages: 930–932 Difficulty: 2 Ans: D

2. One distinction between peptide and steroid hormones is that peptide hormones:

   A) act through nonspecific receptors, whereas steroid hormones act through specific receptors.
   B) are generally water-insoluble, whereas steroid hormones are water soluble.
   C) are more stable than steroid hormones.
   D) bind to cell surface receptors, whereas steroid hormones bind to nuclear receptors.
   E) bind to their receptors with high affinity, whereas steroid hormones bind with low affinity.

Hormones: diverse structures for diverse functions
Page: 932 Difficulty: 2 Ans: D

3. Insulin is an example of a(n) _____ hormone.

   A) catecholamine
   B) eicosanoid
   C) paracrine
   D) peptide
   E) steroid

Hormones: diverse structures for diverse functions
Page: 934 Difficulty: 1 Ans: D

4. The maturation of insulin from its precursor (preproinsulin) involves:

   A) acetylation.
   B) oxidation.
   C) phosphorylation.
   D) proteolysis.
   E) reduction.

Hormones: diverse structures for diverse functions

Page: 934 Difficulty: 2 Ans: D

5. Epinephrine is an example of a(n) _____ hormone.

   A) catecholamine
   B) eicosanoid
   C) paracrine
   D) peptide
   E) steroid

Hormones: diverse structures for diverse functions
Pages: 934–935 Difficulty: 1 Ans: A

6. An example of an eicosanoid hormone is:

   A) epinephrine.
   B) retinoic acid.
   C) testosterone.
   D) thromboxane.
   E) thyroxine.

Hormones: diverse structures for diverse functions
Page: 935 Difficulty: 2 Ans: D

7. An example of a steroid hormone is:

   A) epinephrine.
   B) retinoic acid.
   C) testosterone.
   D) thromboxane.
   E) thyroxine.

Hormones: diverse structures for diverse functions
Page: 935 Difficulty: 2 Ans: C

8. Some hormones are derived from amino acids; for example, catecholamines are derived from _____ while NO is derived from _____.

   A) tyrosine; arginine
   B) tryptophan; lysine
   C) tyrosine; histidine
   D) tryptophan; arginine
   E) histidine; lysine

Hormones: diverse structures for diverse functions
Pages: 934–936 Difficulty: 2 Ans: A

9. The tropic hormones (such as thyrotropin, somatotropin, and luteinizing hormone) are produced and released by the:

   A) anterior pituitary.

B) hypothalamus.
C) ovaries.
D) pancreas.
E) posterior pituitary.

Hormones: diverse structures for diverse functions
Pages: 936–937 Difficulty: 2 Ans: A

10. The normal sequence of action of these components of the hormonal hierarchy is:

A) adrenal cortex → hypothalamus → anterior pituitary.
B) anterior pituitary → adrenal cortex → hypothalamus.
C) anterior pituitary → hypothalamus → adrenal cortex.
D) hypothalamus → adrenal cortex → anterior pituitary.
E) hypothalamus → anterior pituitary → adrenal cortex.

Hormones: diverse structures for diverse functions
Pages: 936–937 Difficulty: 2 Ans: E

11. In its role in the hormonal hierarchy, the hypothalamus produces and releases:

A) epinephrine.
B) insulin.
C) progesterone.
D) releasing factors.
E) thyroxine.

Hormones: diverse structures for diverse functions
Pages: 936–937 Difficulty: 2 Ans: D

12. Which of the following statements about metabolism in the mammalian liver is *false*?

A) Most plasma lipoproteins are synthesized in the liver.
B) The enzymatic complement of liver tissue changes in response to changes in the diet.
C) The liver synthesizes most of the urea produced in the body.
D) The presence of glucose 6-phosphatase makes liver uniquely able to release glucose into the bloodstream.
E) Under certain conditions, most of the functions of the liver can be performed by other organs.

Tissue-specific metabolism: the division of labor
Pages: 939–943 Difficulty: 2 Ans: E

13. Glucokinase:

A) acts in the conversion of liver glycogen to glucose 1-phosphate.
B) converts fructose-6-phosphate to glucose-6-phosphate
C) converts glucose 6-phosphate to fructose 6-phosphate.
D) is a hexokinase isozyme found in liver hepatocytes.
E) is found in all mammalian tissues.

Tissue-specific metabolism: the division of labor
Page: 940 Difficulty: 2 Ans: D

14. Which of the following is *not* a fate of amino acids in the liver?

A) Amino acids are incorporated into new proteins.
B) Nitrogen equivalents from amino acids are converted to urea.
C) Carbon equivalents from amino acids are converted to glucose.
D) Amino acids are stored for later use.
E) Amino acids serve as precursors to nucleotide biosynthesis.

Tissue-specific metabolism: the division of labor
Page: 941 Difficulty: 2 Ans: D

15. Which of the following is *not* a fate of fatty acids in the liver?

A) Fatty acids are oxidized to generate ATP.
B) Fatty acids are converted to ketone bodies.
C) Fatty acids can be converted to cholesterol.
D) Fatty acids are stored for later use.
E) Fatty acids are converted to glucose.

Tissue-specific metabolism: the division of labor
Pages: 942–943 Difficulty: 2 Ans: E

16. A major difference between white adipose tissue and brown adipose tissue is that only:

A) white adipose tissue stores fatty acids as triacylglycerols.
B) brown adipose tissue degrades fatty acids via the citric acid cycle.
C) brown adipose tissue contains thermogenin.
D) white adipose tissue contains adipocytes.
E) white adipose tissue responds to hormone signals.

Tissue-specific metabolism: the division of labor
Pages: 943–944 Difficulty: 2 Ans: C

17. In skeletal muscle:

A) amino acids are an essential fuel.
B) at rest, fatty acids are the preferred fuel.
C) large quantities of triacylglycerol are stored as fuel.
D) phosphocreatine can substitute for ATP as the direct source of energy for muscle contraction.
E) stored muscle glycogen can be converted to glucose and released to replenish blood glucose.

Tissue-specific metabolism: the division of labor
Pages: 944–948 Difficulty: 2 Ans: B

18. In skeletal muscle, phosphocreatine functions as:

A) a reservoir of P_i for mitochondria.

B) reservoir of high-energy of phosphate to replenish ATP.
C) reservoir of amino acids for protein synthesis.
D) an electron acceptor under anaerobic conditions.
E) None of the above

Tissue-specific metabolism: the division of labor
Pages: 944–948 Difficulty: 2 Ans: B

19. In anaerobic skeletal muscle, breakdown of glycogen yields ____ ATP per glucose equivalent.

A) 1
B) 2
C) 3
D) 4
E) 32

Tissue-specific metabolism: the division of labor
Pages: 945–946 Difficulty: 2 Ans: C

20. The Cori cycle is:

A) the conversion of lactate to pyruvate in skeletal muscle to drive glycogen synthesis.
B) the interconversion between glycogen and glucose l-phosphate.
C) the production of lactate from glucose in peripheral tissues with the resynthesis of glucose from lactate in liver.
D) the synthesis of alanine from pyruvate in skeletal muscle and the synthesis of pyruvate from alanine in liver.
E) the synthesis of urea in liver and degradation of urea to carbon dioxide and ammonia by bacteria in the gut.

Tissue-specific metabolism: the division of labor
Pages: 947–948 Difficulty: 2 Ans: C

21. Which one of the following statements is *true*?

A) The brain prefers glucose as an energy source, but can use ketone bodies.
B) Muscle cannot use fatty acids as an energy source.
C) In a well-fed human, about equal amounts of energy are stored as glycogen and as triacylglycerol.
D) Fatty acids cannot be used as an energy source in humans because humans lack the enzymes of the glyoxylate cycle.
E) Amino acids are a preferable energy source over fatty acids.

Tissue-specific metabolism: the division of labor
Page: 949 Difficulty: 2 Ans: A

22. When blood glucose is abnormally high, the pancreas releases:

A) epinephrine.
B) glucagon.
C) glucose.
D) insulin.

E) trypsin.

Hormonal regulation of fuel metabolism
Pages: 951–953 Difficulty: 1 Ans: D

23. When blood glucose is abnormally low, the pancreas releases:

A) epinephrine.
B) glucagon.
C) glucose.
D) insulin.
E) trypsin.

Hormonal regulation of fuel metabolism
Pages: 955–956 Difficulty: 1 Ans: B

24. An elevated insulin level in the blood:

A) inhibits glucose uptake by the liver.
B) inhibits glycogen synthesis in the liver and muscle.
C) results from a below-normal blood glucose level.
D) stimulates glycogen breakdown in liver.
E) stimulates synthesis of fatty acids and triacylglycerols in the liver.

Hormonal regulation of fuel metabolism
Pages: 951–953 Difficulty: 2 Ans: E

25. Which of the following is *not* involved in triggering insulin from β-cells release in response to glucose?

A) Reduced efflux through ATP-gated K^+-channels
B) Increased concentrations of ATP
C) cAMP-dependent phosphorylation of lipase
D) Phosphorylation of glucose to glucose-6-phosphate
E) Increased Ca^{2+} levels

Hormonal regulation of fuel metabolism
Pages: 953–954 Difficulty: 2 Ans: C

26. The largest energy store in a well-nourished human is:

A) ATP in all tissues.
B) blood glucose.
C) liver glycogen.
D) muscle glycogen.
E) triacylglycerols in adipose tissue.

Hormonal regulation of fuel metabolism
Pages: 956–957 Difficulty: 2 Ans: E

27. Elevated epinephrine levels do *not* normally stimulate:

A) fatty acid mobilization in adipose tissue.
B) gluconeogenesis in liver.
C) glycogen breakdown in muscle.
D) glycogen synthesis in liver.
E) glycolysis in muscle.

Hormonal regulation of fuel metabolism
Pages: 958–959 Difficulty: 2 Ans: D

28. Epinephrine triggers an increased rate of glycolysis in muscle by:

A) activation of hexokinase.
B) activation of phosphofructokinase-1.
C) conversion of glycogen phosphorylase a to glycogen phosphorylase b.
D) inhibition of the Cori cycle
E) the Pasteur effect.

Hormonal regulation of fuel metabolism
Pages: 958–959 Difficulty: 3 Ans: B

29. Long-term maintenance of body weight is regulated by the hormone:

A) adiposin.
B) hypothalmin.
C) leptin.
D) obesin.
E) testosterone.

Hormonal regulation of fuel metabolism
Pages: 961–963 Difficulty: 2 Ans: C

30. Which of the following effects is *not* mediated by the protein leptin?

A) Increased synthesis of uncoupler protein thermogenin
B) Decreased activity of the enzyme 5′-AMP-activated protein kinase (AMPK)
C) Increased activity of the sympathetic nervous system
D) Increased β-oxidation
E) Decreased desire to eat

Obesity and the regulation of body mass
Pages: 961–963 Difficulty: 2 Ans: B

31. The hormone leptin _____ appetite; insulin _____ appetite.

A) increases; increases
B) increases; decreases
C) decreases; increases
D) decreases; decreases
E) has no effect on; has no effect on

Obesity and the regulation of body mass

Pages: 962–964 Difficulty: 1 Ans: D

32. The peptide hormone adiponectin, produced in adipose tissue, circulates in the blood and:

A) enhances fatty acid synthesis in liver cells.
B) increases the rate of β-oxidation of fatty acids in muscle cells.
C) inhibits glucose uptake and catabolism in muscle and liver cells.
D) reduces the transport of fatty acids into muscle cells.
E) stimulates gluconeogenesis in liver cells.

Obesity and the regulation of body mass
Page: 964 Difficulty: 2 Ans: B

33. Which of the following is *not* a peptide hormone?

A) Grehlin
B) Insulin
C) Adiponectin
D) PPARγ
E) $PYY_{3\text{-}36}$

Obesity and the regulation of body mass
Pages: 964–967 Difficulty: 2 Ans: D

34. Which of the following hormones stimulates appetite?

A) Grehlin
B) Insulin
C) Adiponectin
D) Both A and B
E) Both A and C

Obesity and the regulation of body mass
Pages: 964–967 Difficulty: 2 Ans: E

35. Which of the following is *not* a mechanism for treating diabetes?

A) Increasing TAG by weight loss
B) Activating AMPK through exercise
C) Activating AMPK by taking metformins
D) Activating PPARγ through rosiglitazone
E) Stimulating insulin secretion by taking sulfonylureas

Obesity, the metabolic syndrome, and type 2 diabetes
Pages: 969–971 Difficulty: 2 Ans: A

Short Answer Questions

36. What is a major problem in isolating a new hormone once a bioassay has been developed?

Ans: Because hormones are very potent, they are often present in tissue in very small amounts. As a result, very large amounts of tissue are needed as the starting material for the isolation of the hormone.

Hormones: diverse structures for diverse functions
Page: 930 Difficulty: 2

37. Name five general classes of hormones and give an example of each.

Ans: (1) Peptide (e.g., insulin, glucagon); (2) catecholamine (e.g., epinephrine); (3) steroid (e.g., testosterone, progesterone); (4) eicosanoids (e.g., prostaglandins, thromboxanes); (5) vitamin D (e.g., 1a,25-dihydroxycalcitrol; (6) retinoid (e.g., retinoic acid); (7) thyroid (e.g., triiodothyronine); (8) nitric oxide (e.g., nitric oxide). (See Table 23-1.)

Hormones: diverse structures for diverse functions
Pages: 933–936 Difficulty: 3

38. Some hormones trigger very rapid responses, whereas for others the response takes much longer to develop. What generalization about the mechanisms of action of these two types of hormones can explain the differences in response times?

Ans: Fast-acting hormones affect the activity of preexisting cellular enzymes. Slow-acting hormones alter gene expression thereby changing the levels of active cellular components.

Hormones: diverse structures for diverse functions
Page: 933 Difficulty: 2

39. Provide some distinguishing characteristics between eicosanoids and hormones such as epinephrine.

Ans: Eicosanoids are paracrine hormones derived from fatty acids: They generally do not move long distances between their points of release and their points of action. Also, eicosanoids are not synthesized in advance but rather are produced as needed. Epinephrine is a member of the chatecholamine hormones and is water soluble. Epinephrine is synthesized in advance, stored in vesicles, and when released, circulates through the bloodstream to remote tissues.

Hormones: diverse structures for diverse functions
Pages: 934–935 Difficulty: 3

40. Provide some distinguishing characteristics between peptide hormones and thyroid hormones.

Ans: Although both peptide hormones and thyroid hormones are composed of amino acids and are subject to proteolytic activation, peptide hormones are generally very small proteins (<100 amino acids) and thyroid hormones are very large proteins (>500 amino acids). Only thyroid hormones are post-translationally modified by iodination of tyrosines.

Hormones: diverse structures for diverse functions
Pages: 934–936 Difficulty: 3

41. Which class of hormones acts via nuclear receptors? Give an example of this type of hormone and briefly describe its mode of action.

Ans: Steroid hormones. Examples are the sex hormones testosterone and estradiol. They pass through the plasma membrane and interact with receptor proteins in the nucleus. The hormone-receptor complex interacts with DNA and alters the expression of specific genes.

Hormones: diverse structures for diverse functions
Page: 935 Difficulty: 2

42. How do hormonal cascades result in large amplification of the original signal?

Ans: At each level, a small quantity of the signal molecule activates a larger number of molecules at the next level. When this occurs over several levels, this multiplicative effect can result in a several million-fold amplification of the original signal. (See Fig. 23-7.)

Hormones: diverse structures for diverse functions
Page: 937 Difficulty: 2

43. Describe five possible fates for glucose 6-phosphate in the liver.

Ans: Possible answers include: (1) conversion to liver glycogen; (2) dephosphorylation and release of glucose into bloodstream; (3) oxidation via the pentose phosphate pathway; (4) oxidation via glycolysis and the citric acid cycle; (5) oxidation to acetyl-CoA, which then serves as precursor for synthesis of triacylglycerols, phospholipids, and cholesterol. (See Fig. 23-12.)

Tissue-specific metabolism: the division of labor
Pages: 940–941 Difficulty: 2

44. Describe five possible fates of amino acids arriving in the liver after intestinal uptake.

Ans: Possible answers include: (1) synthesis of nucleotides; (2) synthesis of hormones; (3) synthesis of other nitrogenous products such as porphyrins; (4) deamination, followed by oxidation of carbon skeleton for energy; (5) synthesis of proteins for export to plasma; (6) synthesis of proteins for liver; (7) export of free amino acids to other tissues. (See Fig. 23-13.)

Tissue-specific metabolism: the division of labor
Pages: 941–942 Difficulty: 2

45. Describe five possible fates for fatty acids in the liver.

Ans: Possible answers include: (1) conversion to triacylglycerol or cholesterol esters for export in plasma lipoproteins; (2) conversion into hepatocyte phospholipids; (3) oxidation and conversion to ketone bodies for export to other tissues; (4) β-oxidation to acetyl-CoA, and further oxidation via citric acid cycle for ATP production; (5) β-oxidation to acetyl-CoA, followed by synthesis of cholesterol from acetyl-CoA; (6) binding to serum albumin for transport to heart and skeleton. (See Fig. 23-14.)

Tissue-specific metabolism: the division of labor
Pages: 942–943 Difficulty: 2

46. Provide three distinguishing characteristics between white adipose tissue (WAT) and brown adipose tissue (BAT).

Ans: Possible answers include: (1) WAT is more abundant than BAT. (2) The adipocytes of WAT are large and completely filled with a single droplet of triacylglycerols. The adipocytes of BAT are smaller and store triacylglycerols in many smaller droplets. (3) In WAT, burning of triacylglycerols leads to ATP formation, whereas in BAT, much of that energy is converted to heat instead. (4) BAT cells have more mitochondria than WAT. (5) BAT has a richer supply of capillaries than WAT.

Tissue-specific metabolism: the division of labor
Pages: 943–944 Difficulty: 2

47. Briefly explain the role of phosphocreatine in muscle.

Ans: During maximal muscle activity, muscle cells are in danger of depleting their ATP levels. Oxygen cannot be provided rapidly enough to support the electron transport chain, fermentation is not nearly as efficient as aerobic metabolism, and the decrease in pH caused by lactate accumulation is also detrimental to the efficiency of the muscles. Phosphocreatine provides an additional source of energy that can convert ADP back to ATP via the creatine kinase reaction to help maintain muscle activity.

Tissue-specific metabolism: the division of labor
Pages: 945–947 Difficulty: 2

48. Briefly explain the Cori cycle.

Ans: The Cori cycle allows for lactate produced by anaerobic muscles to be transported to the liver where it is converted back to glucose. The glucose is then transported back to the muscle to complete the cycle.

Tissue-specific metabolism: the division of labor
Pages: 947–948 Difficulty: 2

49. Compare in general terms the effects of epinephrine, glucagon, and insulin on glucose metabolism.

Ans: Epinephrine and glucagon cause an increase in the blood glucose level. Epinephrine acts when a higher than normal level of glucose is required; glucagon acts when the level is unusually low. Both stimulate gluconeogenesis and glycogen breakdown and decrease glycolysis and glycogen synthesis. Insulin causes a decrease in blood glucose levels; it acts by increasing glycogen synthesis, glycolysis, and glucose uptake by cells as well as by decreasing glycogen breakdown.

Hormonal regulation of fuel metabolism
Pages: 951–958 Difficulty: 2

50. Explain the signaling pathway by which an increase in blood glucose causes the release of insulin from pancreatic β-cells.

Ans: Glucose in the blood is transported into pancreatic β-cells via the GLUT2 transporter, where it is phosphorylated and then metabolized via glycolysis and the citric acid cycle, ultimately leading to higher ATP concentrations via oxidative phosphorylation. Higher ATP levels close the ATP-gated K^+ channels, depolarizing the membrane, which leads to the opening of the voltage-gated Ca^{2+} channels, an increase in intracellular Ca^{2+} concentrations, and the subsequent release of insulin via exocytosis.

Hormonal regulation of fuel metabolism

Pages: 953–954 Difficulty: 2

51. Explain the response and the consequences of prolonged fasting on fuel metabolism by the liver.

Ans: Under conditions of prolonged fasting (i.e., absence of glucose in the diet), the liver is responsible for providing energy, primarily to the brain. Fatty acids from adipocytes are broken down and converted to ketone bodies, which can be used as an alternative fuel source by the brain. Proteins from muscle cells are broken down to provide an alternative route to glucose via citric acid cycle intermediates and gluconeogenesis. The consequences include muscle wastage, loss of weight, acidic blood, and excess ketone bodies and urea in the urine.

Hormonal regulation of fuel metabolism
Pages: 956–958 Difficulty: 2

52. What is leptin? How does it function in the long-term maintenance of body mass?

Ans: Leptin is a small polypeptide that is produced in adipocytes and is carried by the blood to the brain. It is produced when there are adequate stores of lipid in the adipocytes and interacts with a receptor in the hypothalamus. This interaction produces a cascade of effects that tend to suppress appetite and increase energy expenditure.

Obesity and the regulation of body mass
Pages: 961–963 Difficulty: 2

53. Describe the signaling cascade initiated by leptin binding to its receptor.

Ans: Binding of leptin to its receptor in the hypothalamus causes the receptor to dimerize. The soluble cytoplasmic "Janus kinase" (JAK) can now bind to receptor, become activated, and phosphorylate tyrosine residues on the receptor's intracellular domain. These residues in turn become docking sites for three of the STAT transcription factors, positioning them for phosphorylation by JAK as well. The phosphorylated STATs then dimerize and can migrate into the nucleus, where they activate transcription of specific genes, including that for proopiomelanocortin, from which α-melanocyte-stimulating hormone (α-MSH) is produced. The signal that α-MSH conveys to the brain is to "stop eating!"

Obesity and the regulation of body mass
Pages: 961–963 Difficulty: 2

54. Briefly describe the effects of the hormone adiponectin.

Ans: Adiponectin is a peptide hormone produced by adipose tissues that triggers increased uptake and oxidation of fatty acids in liver and muscle cells. It also blocks gluconeogenesis and fatty acid biosynthesis in the liver, and sensitizes the liver to insulin.

Obesity and the regulation of body mass
Pages: 964–965 Difficulty: 2

55. Briefly describe the "lipid burden" hypothesis that explains the development of type 2 diabetes.

Ans: Excessive caloric intake by obese individuals causes adipocytes to become completely filled with

triacyclglycerols, unable to take up and store any additional fatty acids. These lipid-filled adipocytes release signals that attract macrophages that invade the fat tissue and trigger an inflammatory response. The inflammatory response further impairs uptake of fatty acids and leads to even higher levels of free fatty acids in the bloodstream. The excess free fatty acids are taken up by the liver and muscle cells. The fatty acids are then converted to triacylglycerols and stored as lipid droplets. These lipid droplets are toxic to these cells, leading to eventual insulin resistance through a variety of mechanisms.

Obesity, the metabolic syndrome, and type 2 diabetes
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